

Map Portfolio

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As built to GIS: City government client had an existing utility GIS and wanted to add new construction project. I was given the new construction as built drawings (examples shown in Figures 1-5). I used Portal in ArcGIS Pro to bring in utility feature layers from ArcGIS Online. I used the Edit tools to create, modify, and update attributes (examples shown in Figures 6-10). These updates were synced to ArcGIS Online and ensured that their city utility data was up-to-date. Utility feature layers were used in public web maps and dashboards. Below are examples of some of the as built and their corresponding GIS feature layers that I created.

Figure 1:

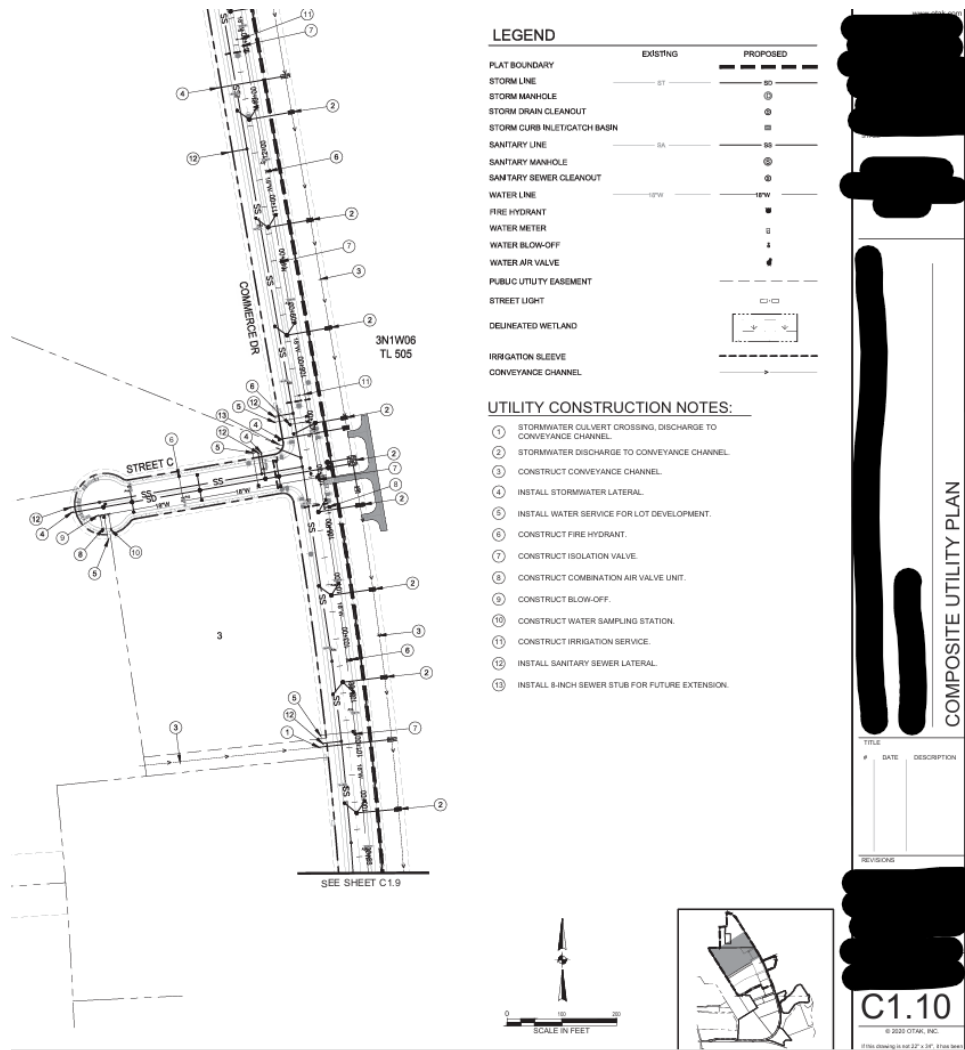


Figure 2:

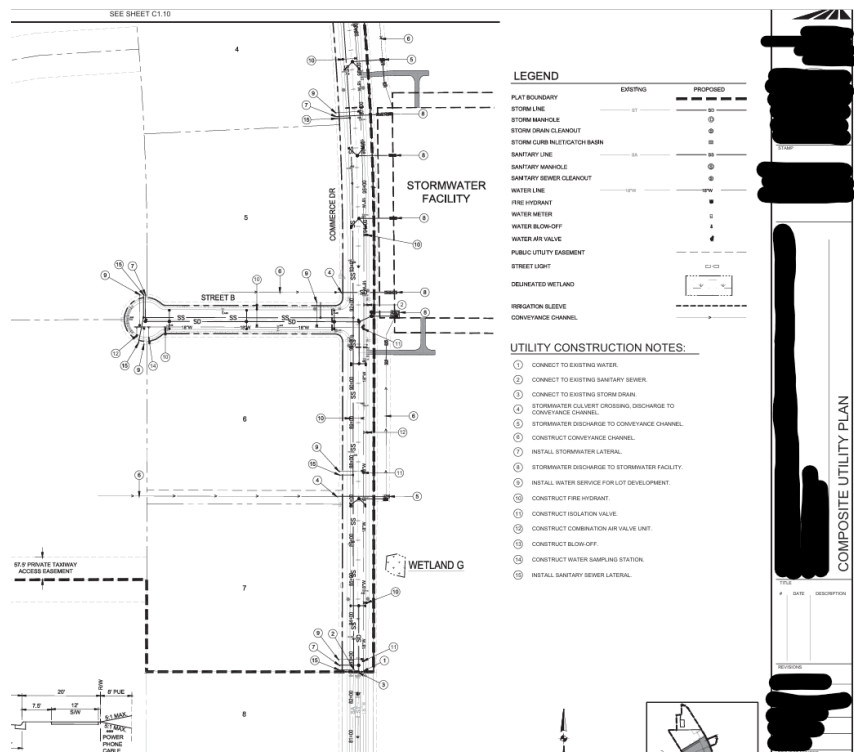
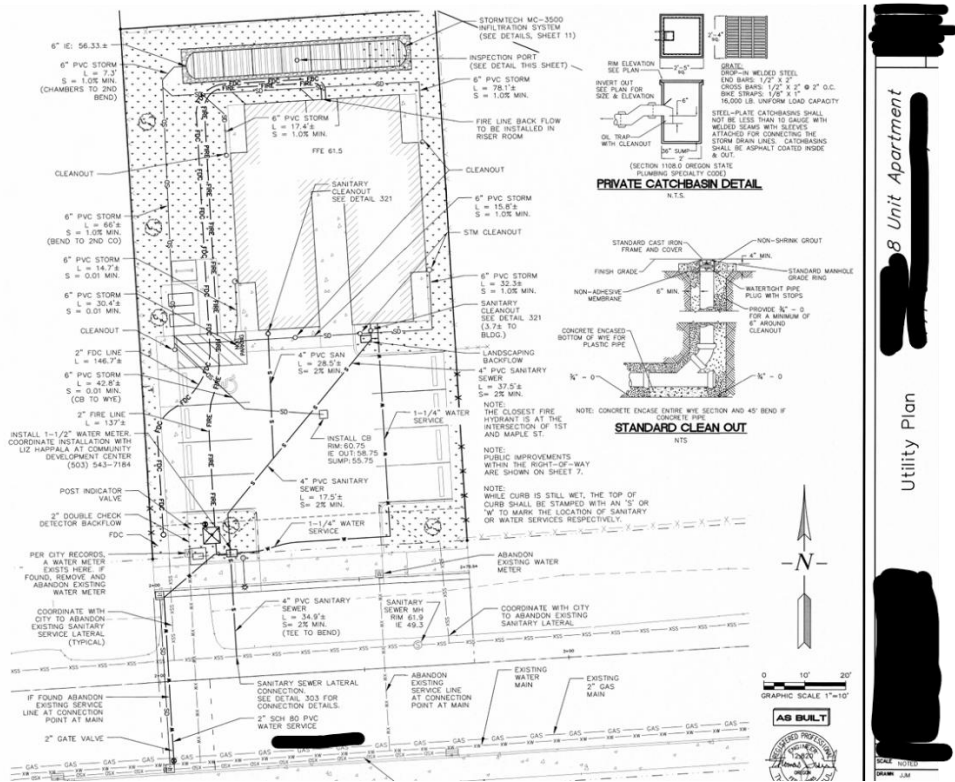


Figure 3:



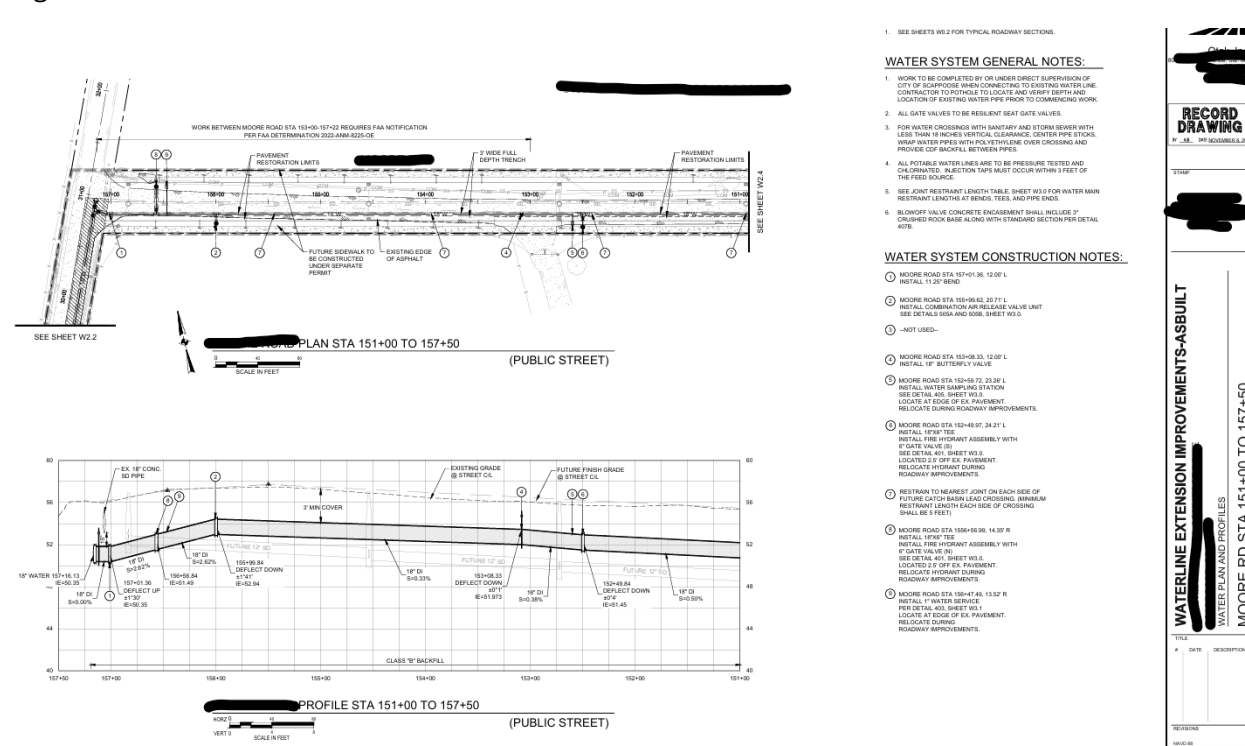


Figure 6:

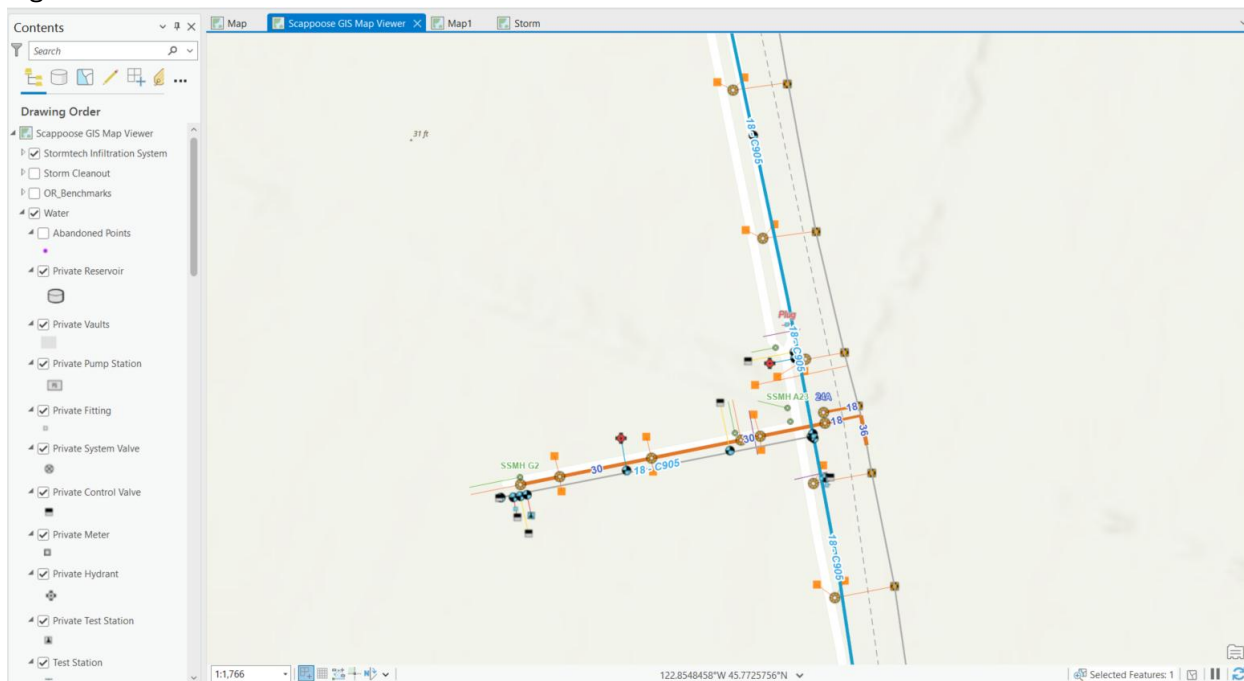


Figure 7:

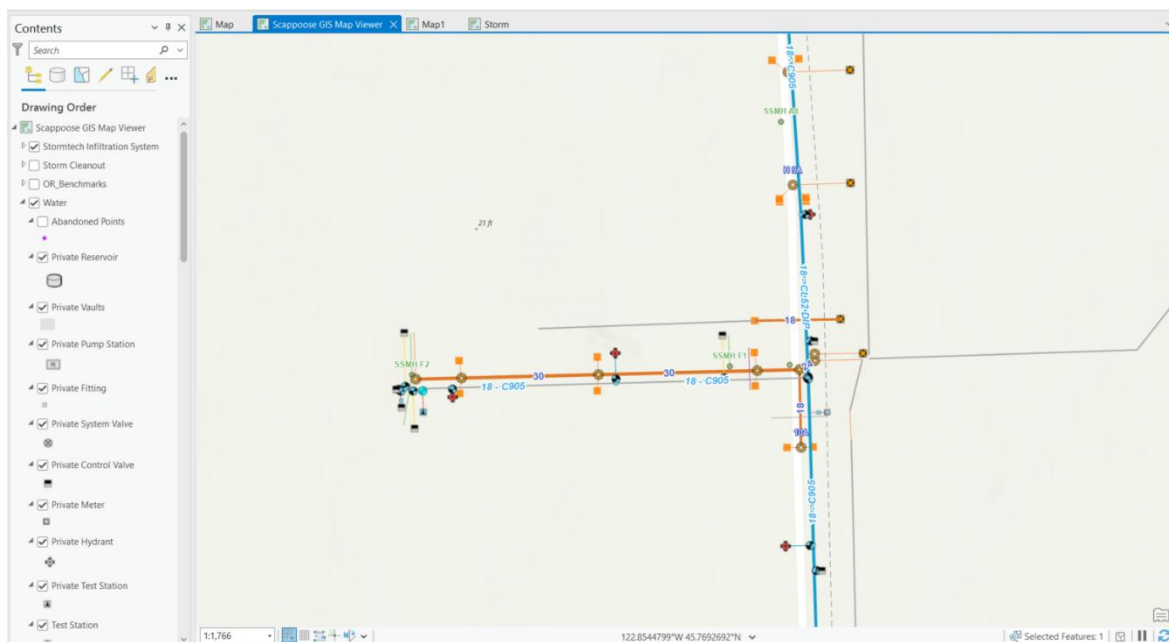


Figure 8:

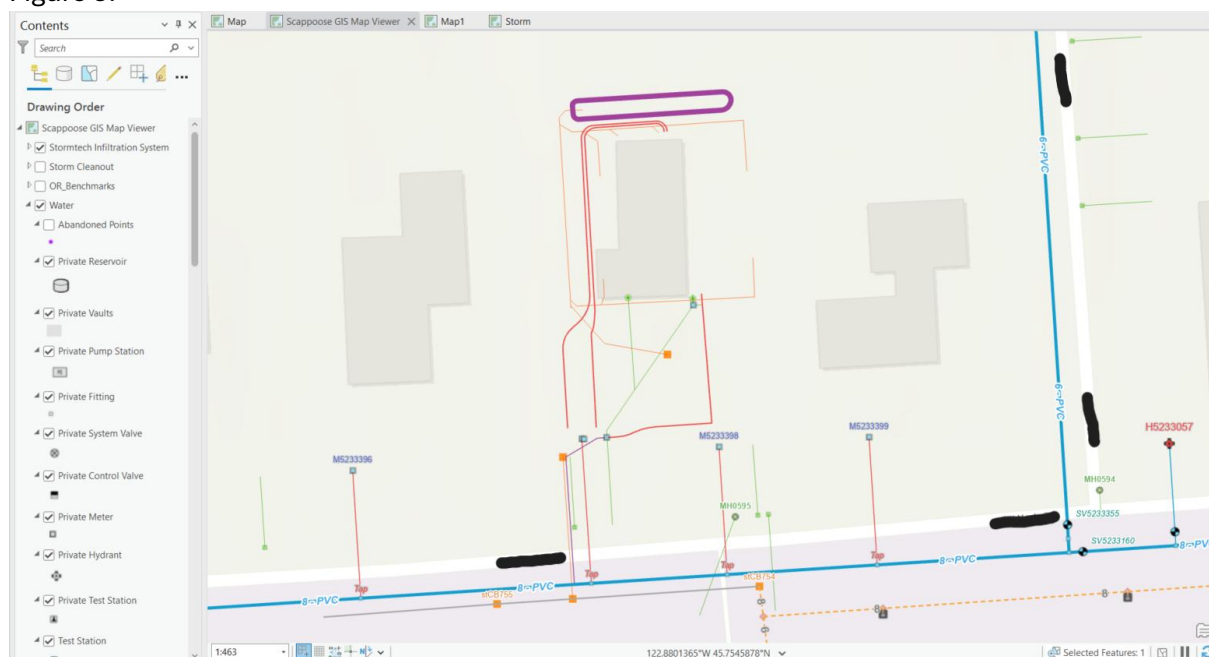


Figure 9:

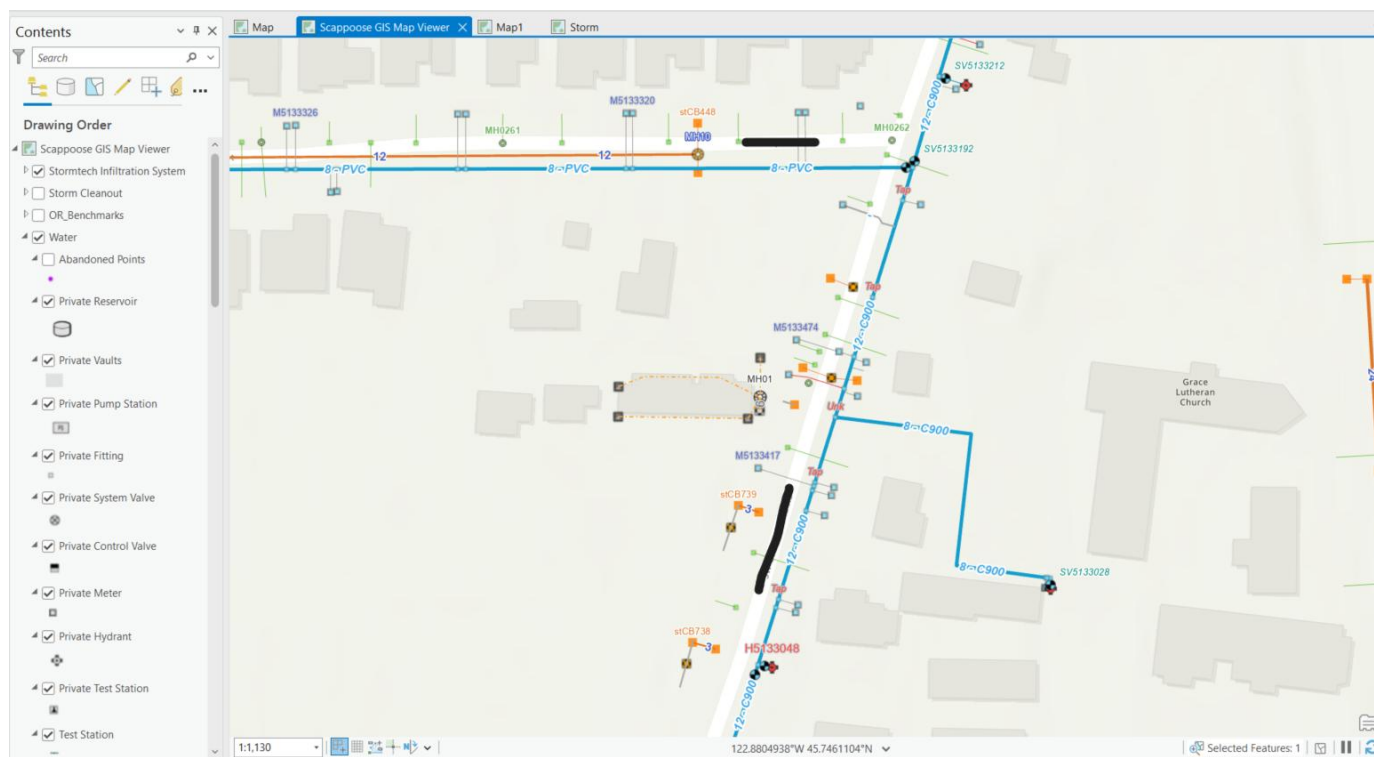
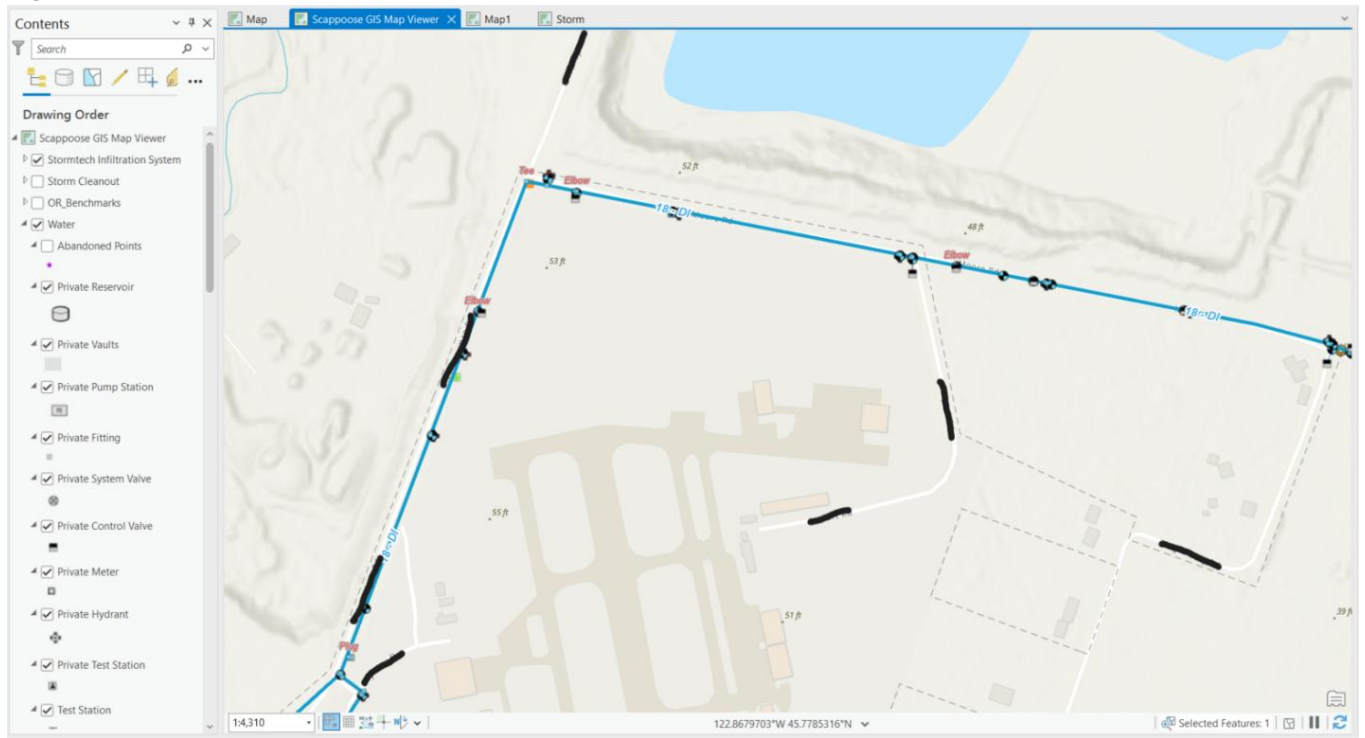


Figure 10:



CSV to GIS: Municipal government utility needed help developing their GIS. They had existing CSV files and as built drawings, but no web maps or GIS data. I used various GIS tools and methods to help them bring in existing data. I deployed ArcGIS Solutions for Stormwater, Sewer, Water, Electric, and Communication networks through the customer's ArcGIS Online. I then used the Portal to bring in the Solutions layers to ArcGIS Pro. I imported the customer's CSV files, which contained Northing and Easting data, into ArcGIS Pro as feature layers (Figure 11). I then used the Append tool to add the existing Solutions layers with the CSV data, which brought over the symbology and fields to the new layers. I also Georeferenced the customer's as built drawings and used the Editing tools to create new layers based on the as built. Updated Solutions layers were then available in ArcGIS Online (Figure 12). I provided training for the customers so that they can use access their GIS utility layers in Field Maps and use high accuracy GNSS receivers to create new points (Figure 13).

Figure 11:

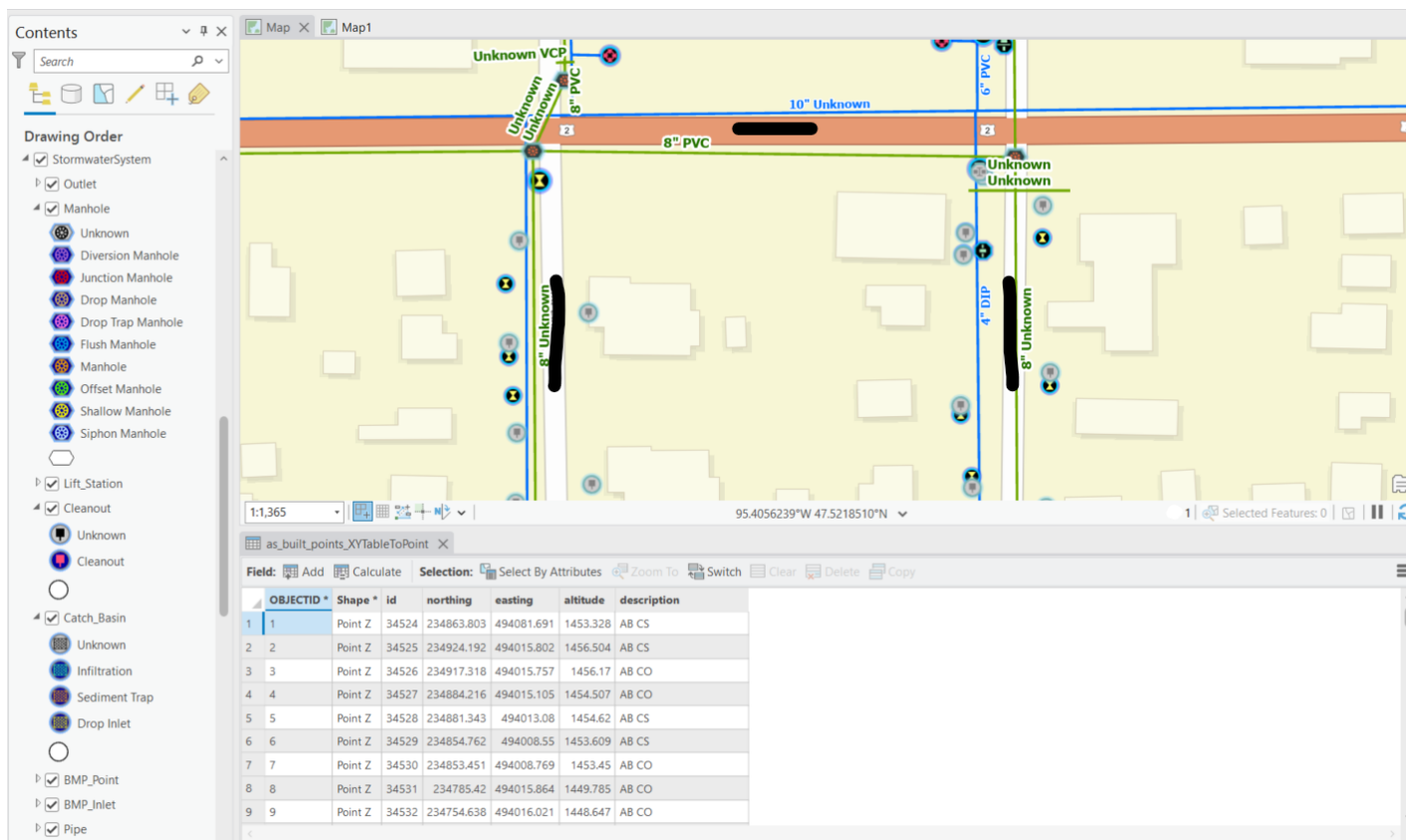


Figure 12:

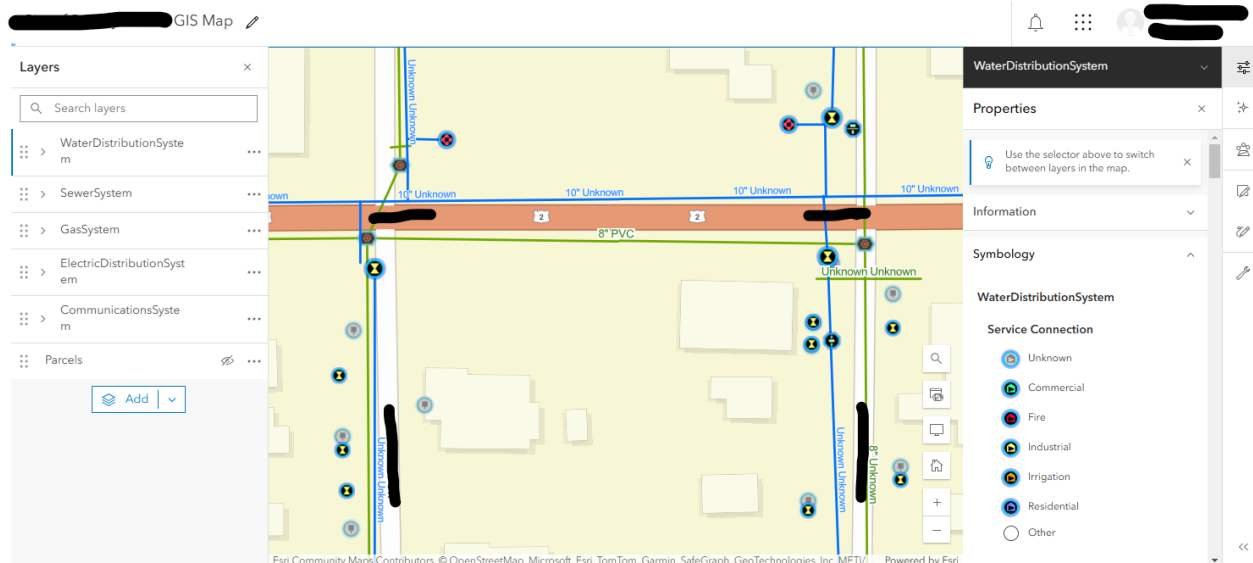
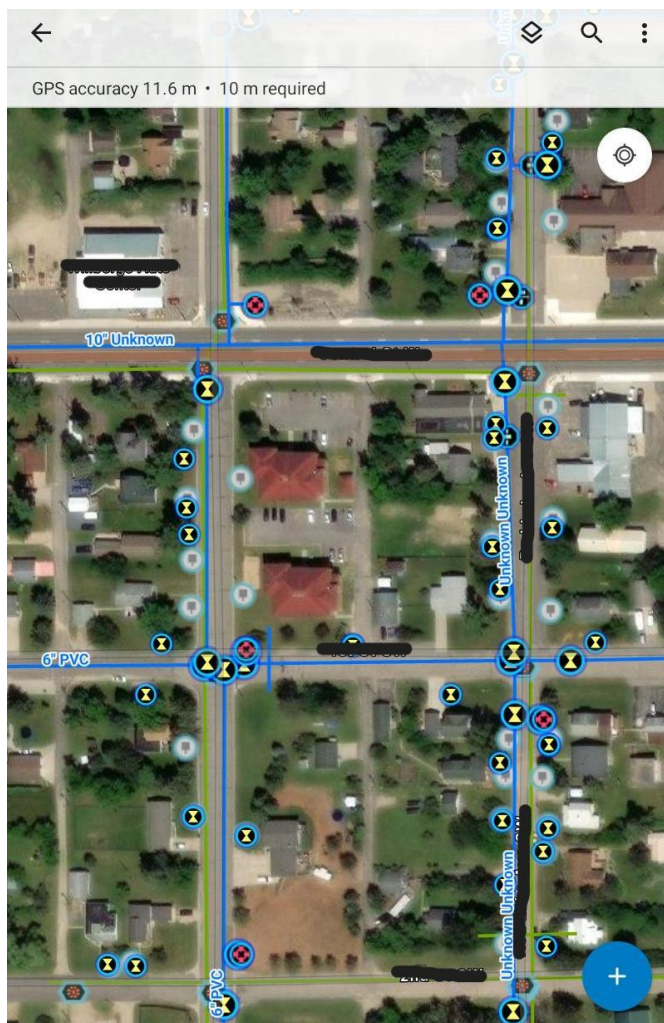
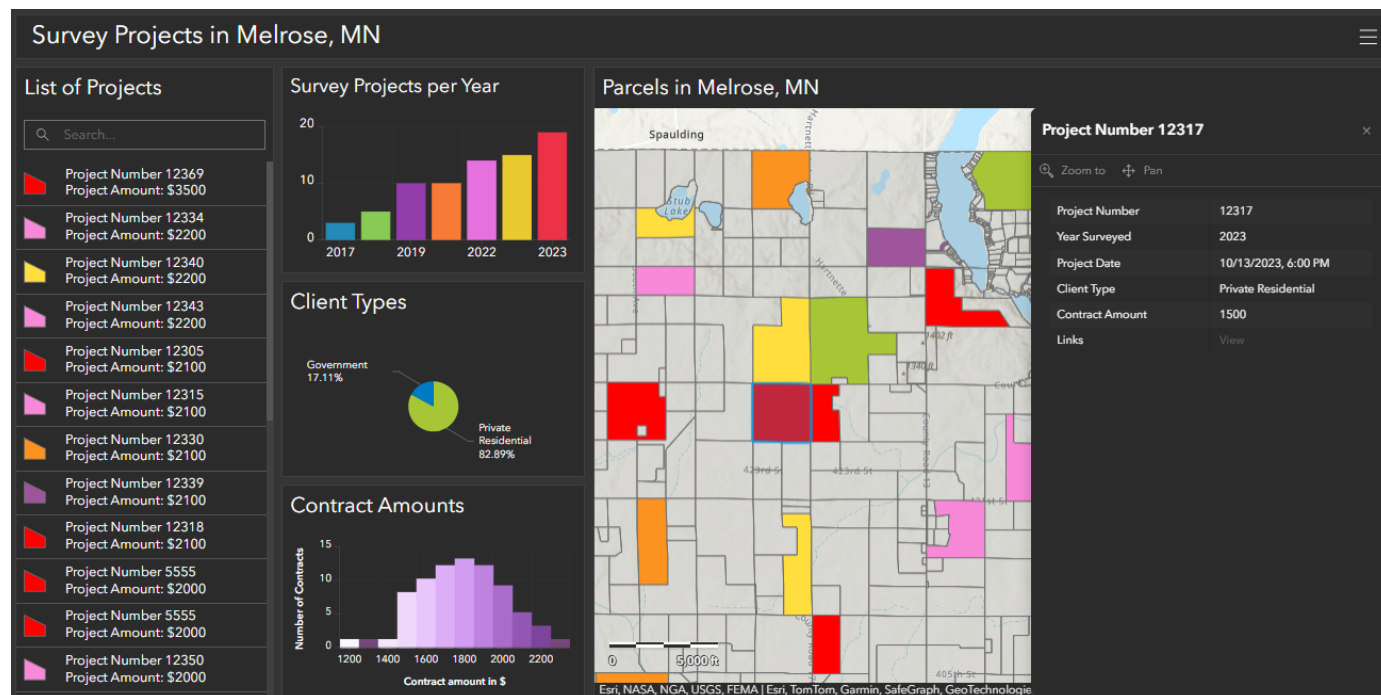


Figure 13:



Survey Data to GIS: I was a panelist in an ESRI webinar for the NSPS called [“I am a Surveyor- How can I use GIS?”](#). For this webinar I used example parcel data that replicated survey projects to create various GIS products. First, I published the data on ArcGIS Online as a hosted feature layer. I used this to create a Web Map, in which I configured the symbology, popups, and labels. I used this map to create a Web App with Instant Apps and configured the app to be compatible on mobile devices. I also created an ESRI Dashboard with the map where I configured a searchable list, charts of field values, and a dynamic web map (Figure 14).

Figure 14:



Species Richness Map: This map visually represents a new data layer I created as part of my master's thesis project (Figure 15). As part of my research, I was building a linear model with county-level data, and I wanted to calculate species richness per county. Simply averaging the number of pixels per county did not accurately capture the variation in species richness, so I had to construct a new method to obtain an accurate species richness count in each county. I did this by first downloading each individual species range map from the USGS GAP database. This database includes 649 bird species range maps at the 30 m pixel resolution. Each range map consists of pixels classified into summer, winter, or year-round habitat. My study uses data for the entire year, so I needed to reclassify the season-specific pixel values to instead be either “present” or “absent” at any point in the year. I used R software (R Core Team, 2022) to reclassify the pixel values to the categorical values of 0 (the species’ range is not present in this pixel) and 1 (the species’ range is present in this pixel). These values were then used to classify each county as 1 or 0. For example, if the county contained any pixels classified as 1 (the species range is present), then the entire county was classified as 1, indicating that the species range is present in that county. These values were calculated for each species in each county, resulting in 649 datasets, one for each bird species, each containing all 3102 counties in the US classified as either 1 or 0. All these datasets were added to calculate a count of the total number of species present in each county. This is the first time that GAP species richness data have been used to calculate bird species richness at the county level for the contiguous US. A county level bird species richness dataset is a helpful addition to the current GAP data available. The current GAP data for birds is only available as a 30m pixel raster for bird species richness in which each pixel represents the number of species that could occur in that pixel, or as 30m pixel rasters for each individual species that represent the potential for the species to occur there in the summer, winter, or year-round. I produced a bird species richness value for each county in the US, which could be used in future human-environment studies that use other county-level data. I then used R to build a linear spatial model to study the relationship between bird species richness and available photosynthetic energy in every county in the conterminous US.

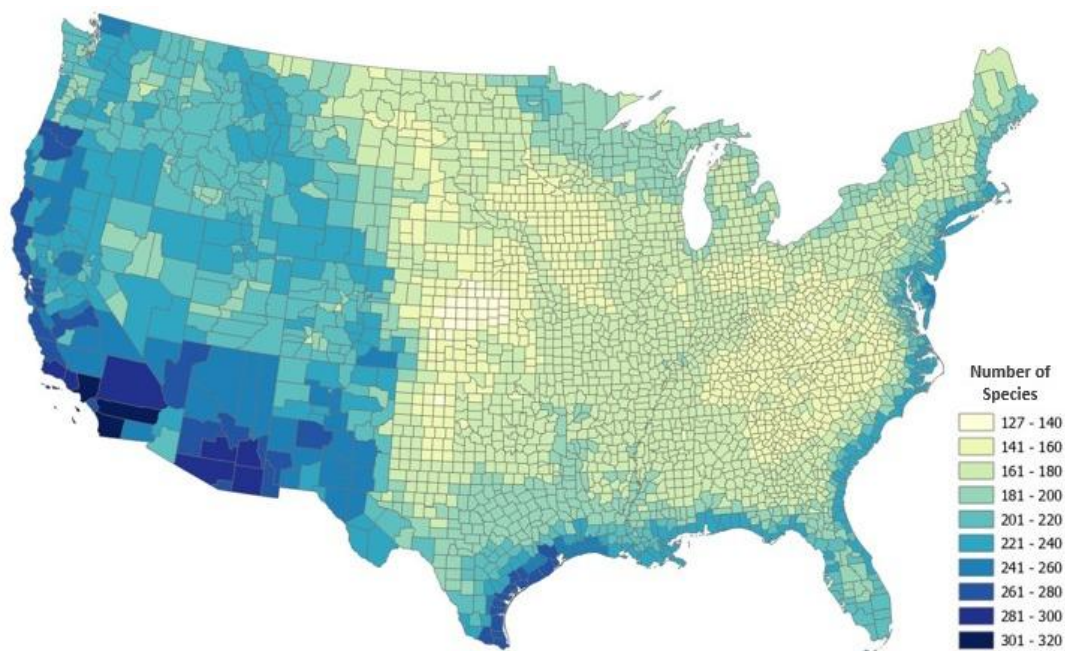


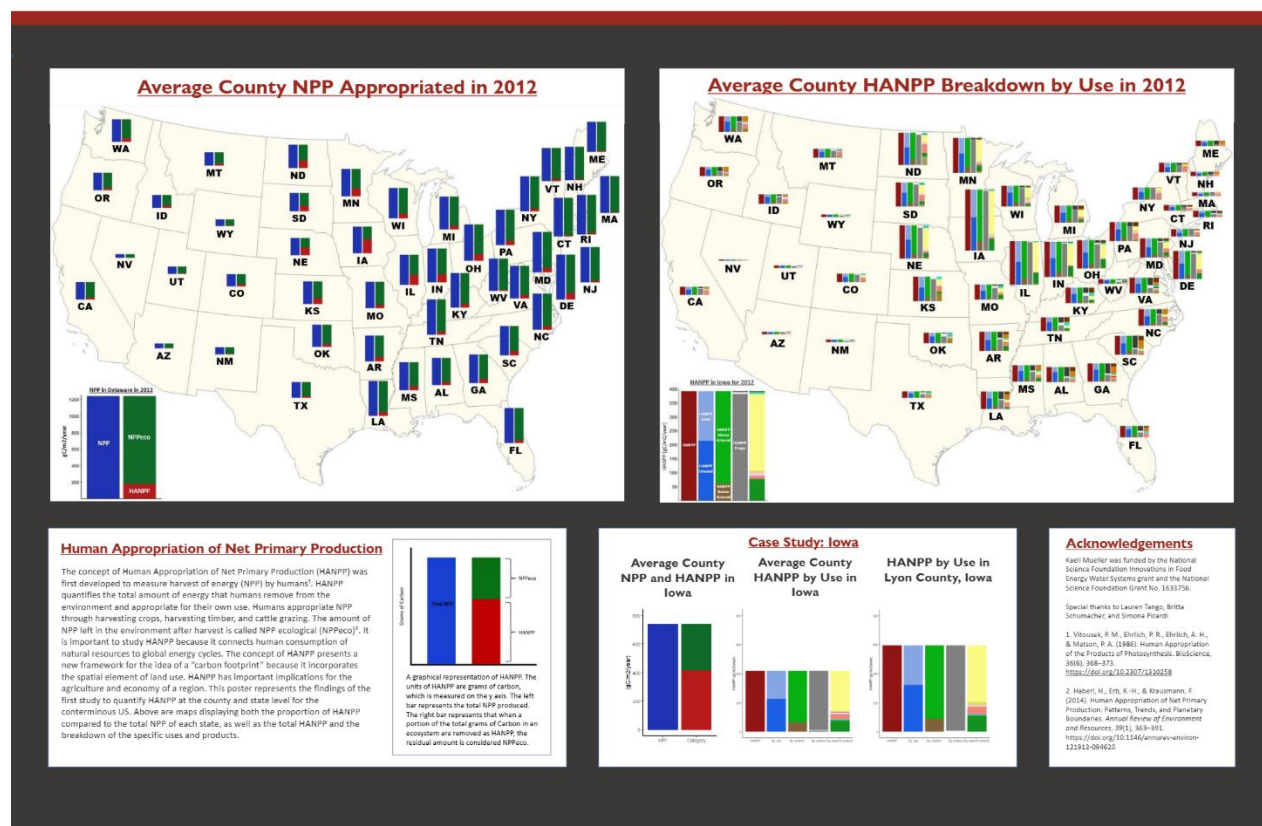
Figure 15:

HANPP State Maps Poster: The concept of Human Appropriation of Net Primary Production (HANPP) was first developed to measure harvest of energy (NPP) by humans. HANPP quantifies the total amount of energy that humans remove from the environment and appropriate for their own use. Humans appropriate NPP through harvesting crops, harvesting timber, and cattle grazing. The amount of NPP left in the environment after harvest is called NPP ecological (NPPEco). It is important to study HANPP because it connects human consumption of natural resources to global energy cycles. The concept of HANPP presents a new framework for the idea of a “carbon footprint” because it incorporates the spatial element of land use. HANPP has important implications for the agriculture and economy of a region. This poster represents the findings of the first study to quantify HANPP at the county and state level for the conterminous US. Above are maps displaying both the proportion of HANPP compared to the total NPP of each state, as well as the total HANPP and the breakdown of the specific uses and products. Total HANPP (harvest) and its components: used and unused, above and below-ground, from crops, timber, and grazing, and by product for the contiguous U.S. in (a) 1997, (b) 2002, (c) 2007, (d) 2012. Individual bar charts were produced using R. This poster contains HANPP data from 2012 in the form of individual NPP and HANPP bar charts for each state, national totals of NPP and HANPP, and example of county HANPP and NPP using Lyon county Iowa. This poster won a “Best of Show” award at the 2022 Utah State University Spring Runoff Conference (Figure 16).

Figure 16:

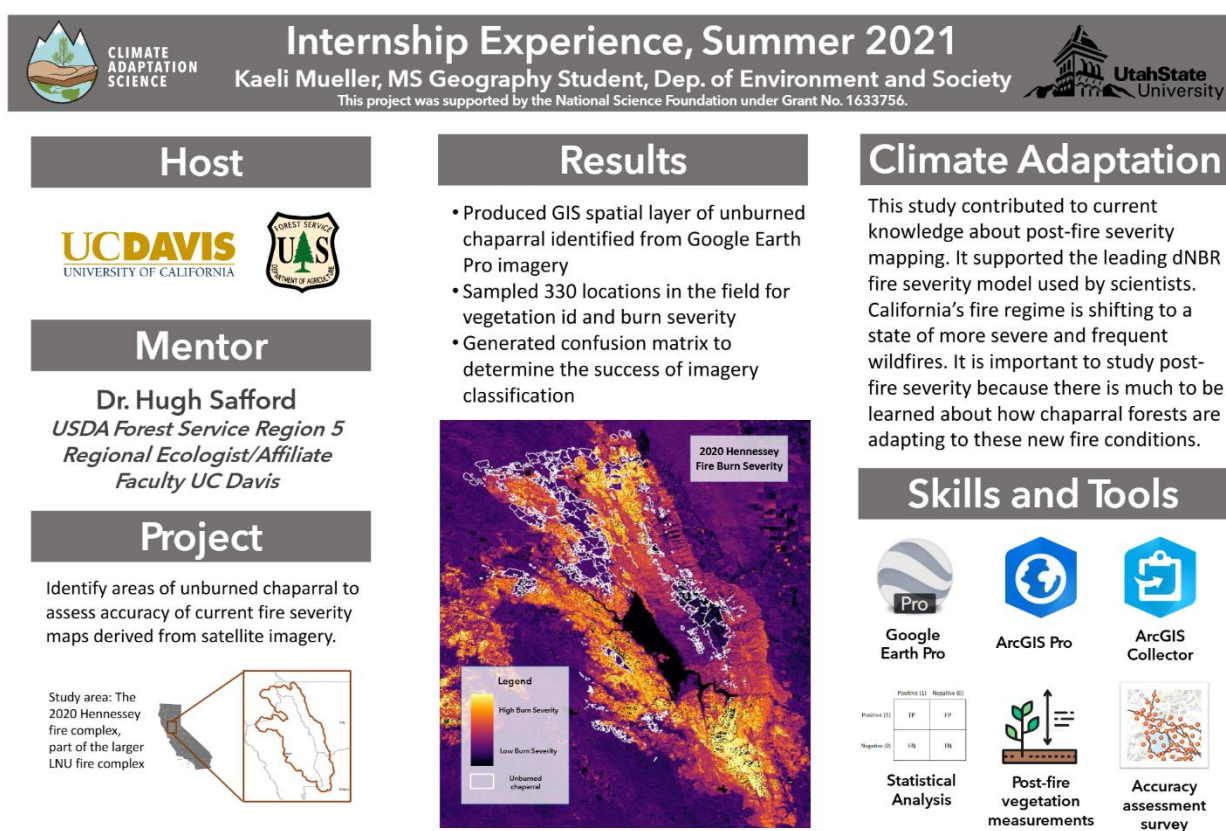
Human Appropriation of Net Primary Production

Kaeli Mueller, Suman Paudel, Christopher Lant, Gustavo Ovando-Montejo



Fire severity map poster: This poster contains a map that I created during my work as a GIS Intern at UC Davis. To make this map, I collected my own data using the ArcGIS Collector app, ArcGIS Online, and Google Earth Pro. I collected polygons of unburned chaparral forest through Google Earth Pro and converted them to ArcGIS-compatible shapefiles. I then collected GPS points of burned and unburned chaparral forest using the ArcGIS Collector App and converted them to shapefile layer in ArcGIS Online. I performed an accuracy assessment of my Google Earth Polygons with my ground truth data to assess the validity of my own fire severity map. I then overlaid this new layer I had created with existing fire severity raster models to visually assess their accuracy. During this experience I had lots of opportunities to problem solve by creating new workflows between different GIS software. I created this poster for my Climate Adaptation Science project and shared it at a Science Management and Policy Exchange Meeting (Figure 17).

Figure 17:



Habitat Suitability Models: These maps represent habitat suitability for 7 tree species of interest across the Sierra Madre Occidental for current climatic conditions, end-of-century (2100) climatic conditions for SSP 3-7.0, and end-of-century (2100) climatic conditions for SSP 5-8.5 (Figure 18). Habitat suitability probability is represented on a gradient of blue (low suitability) to red (high suitability). Predicted model probability range is in the upper right corner of each species map as [lower-upper%]. Figure column is assigned numerically 337 while column is assigned alphabetically for rapid lookup (i.e., *Pinus Stroboformis* at end-of-century conditions for SSP 3-7.0 = Figure 4.4.B). This figure is part of a paper that is currently under review and is written by myself, Josh Carrell, and April Phinney. The study explores the potential effects of climate change on the habitat suitability of seven tree species at two distinct spatial scales: the Coronado National Forest (CNF), a local management area, and the Sierra Madre Occidental (SMO), an ecoregional extent. Habitat suitability was determined by extrapolating Ecological Niche Models based on citizen-science tree occurrence records into future climatic conditions using projected 30-year normals for two anthropogenic emissions scenarios through the end of the century. Habitat suitability for seven common tree species was modeled using three machine learning algorithms commonly applied in ENM for current climatic conditions and 2 future anthropogenic emissions scenarios across 3 distinct 30-year averages. Mean ensemble models were generated from the spatial outputs of the four algorithms for each of the seven scenarios. In Ecological Niche Modelling, no one model is constituted as “best”. Therefore, we implemented a mean ensemble approach from three machine learning algorithms commonly applied in ENM. These models include: 1) Random Forest; 2) Maximum Entropy; 3) Boosted Regression Trees. Models were fit with 38 species-specific climatic, topographic, and edaphic variables that reflect common drivers of plant distribution as well as species presence data gathered from commonly used citizen-science databases (e.g., iNaturalist). Maximum kappa classification threshold was implemented for consistency across statistical algorithm selection. Accuracy was measured using the Area Under the Curve (AUC), Sensitivity, Specificity, and the True Skill Statistic (TSS). These metrics measure the ability of the models to correctly classify the presence and pseudo-absences in the training data. Each mean ensemble model predicts on the [0,1] interval that represents the estimated probability of species occurrence (i.e., habitat suitability). The results show that habitat suitability is expected to decrease for all seven tree species at varying degrees.

Figure 18:

